Ashley Nand

Gen AI Developer, Android Developer

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Results-driven software engineer with **3+ years of mobile development experience** in **Java** and **Kotlin**, combined with strong expertise in architecting modern Android applications. Skilled in building **AI and GenAI solutions**, including **RAG pipelines**, **multi-agent systems**, and intelligent automation using **Python, LangChain, LlamaIndex**, and leading **LLMs**. Experienced in prompt engineering, retrieval optimization, and developing scalable AI workflows.

Skills

Languages	Java, Kotlin, Python, C++, C, TypeScript, JavaScript, SQL
Backend Development	Spring, Spring MVC, Spring Boot, Node.js, Express.js, RESTful APIs
Mobile Development	Android Development: Java, Kotlin, XML, Jetpack, Room, SQLite, Firebase
Machine Learning & AI	Machine Learning: Scikit-learn, NumPy, Pandas, PyTorch; Generative AI: Large Language Models (LLMs), Transformer Architecture, Prompt Engineering
AI Frameworks	LangChain, LlamaIndex, Streamlit
RAG Systems	Retrieval-Augmented Generation (RAG), Document Q&A Pipelines
AI Agents & Deployment	PydanticAI, AutoGen, CrewAI, Multi-Agent Systems, Model Context Protocol (MCP), Ollama, Docker, vLLM, AWS EC2, OpenAI API, Gemini API, Groq API
Frontend Development	React.js, HTML5, CSS3, Bootstrap, Tailwind CSS, XML
Databases	MySQL, MongoDB, SQLite, Room, Firebase Realtime Database
Tools & DevOps	Android Studio, IntelliJ, PyCharm, Git, GitHub, Postman, Docker, Firebase, Ubuntu Server, CI/CD, VS Code

Technical Experience

Software Development Engineer (F2)
Sep 2023 — Present
Mindsprint (Olam Groups) *Chennai, India*

Generative AI / Agentic AI Engineer *Apr 2024 — Present*
Responsibilities:

- Designed and implemented LLM-powered applications using GPT models, vector databases, and RAG architectures to automate knowledge retrieval and enhance operational efficiency.
- Built and deployed production-grade AI agents for data extraction, task automation, and workflow orchestration using Python, LangChain, FastAPI, and AWS cloud services.
- Built and tested multi-agent workflows using CrewAI and LangChain, along with a Trader Analyst tool for stock market analysis and trend prediction; containerized projects with Docker and deployed on AWS EC2 as part of hands-on learning.
- Improved model performance through prompt engineering, fine-tuning, dataset curation, and continuous monitoring of accuracy, latency, and cost metrics across deployed AI services.

OFIS Enumerator (Published on Play Store) *Sep 2023 — Mar 2024*
Responsibilities:

- Developed and launched the OFIS Enumerator Android app for **10+ million farmers** using Java, XML, Jetpack, and Retrofit with Spring Boot APIs for secure, real-time data synchronization.
- Built offline-first architecture with Room and SQLite, improving reliability in low-connectivity regions and reducing crash rates by **40%**.
- Enhanced performance and delivery using Firebase notifications, crash analytics, and optimized CI/CD pipelines with Git, Jenkins, and Gradle for production deployment on Google Play Store.

TRAINEE – Android Development / OFIS Enumerator (Published on Play Store)
Feb 2023 — Sep 2023 Olam
Information Services Private Limited *Chennai, India*

- Built and maintained Android applications using Java, XML, Room Database, and SQLite, integrating RESTful APIs and improving performance, offline reliability, and overall code quality.
- Led end-to-end development and issue resolution for multiple apps, including publishing the OFIS Enumerator app on the Google Play Store, while managing version control with Git.

Publications

Bi-Directional LSTM Classification of Emotions in EEG Signals **Published: Jan 16, 2023 - IEEE Xplore**

2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), IEEE Xplore. Achieved 94.12% accuracy using Bi-LSTM on DEAP EEG dataset to classify emotions (valence, arousal, dominance) from a single-channel electrode.

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Education

Master of Technology in Artificial Intelligence & Data Science (CGPA: 8.9) *April 2023*
Karunya Institute of Technology & Science

Bachelor of Technology in Computer Science & Engineering (CGPA: 7.9) *August 2021*
Jawaharlal Nehru Technological University, Kakinada

Higher Secondary Education (CBSE) – PCM (CGPA: 7.2) *May 2017*
Kendriya Vidyalaya, Bacheli

Projects

Kofi: Study Assistant using CrewAI, RAG, Docker, EC2 *Oct 2025 – Nov 2025*
Technologies: Python, CrewAI, LlamaIndex, ChromaDB, Streamlit, FastAPI, Groq, Docker, AWS EC2
Built a RAG-based PDF question-answering system using LlamaIndex and ChromaDB for vector search. Developed a FastAPI + CrewAI backend and a Streamlit frontend, leveraging Groq for fast inference. Containerized with Docker and deployed on AWS EC2, optimizing prompts and embeddings for accurate, context-aware results.

The Bible Stories *Nov 2024 – Dec 2024*
Technologies: MySQL, React, Express, Node.js, Tailwind CSS, Ubuntu, GitHub
Developed a full-stack web application for managing and publishing Bible-related content. Implemented secure user authentication, optimized back-end APIs, and orchestrated CI/CD pipelines for robust deployment and scalability.

Time Series Data Analysis – The University of Auckland, New Zealand *Nov 2021 – May 2023*
Technologies: Machine Learning, Python, Big Data
Performed time-series analysis using advanced Python libraries under the mentorship of Dr. Andreas W. Kempa-Liehr. Explored modeling, forecasting, and evaluation techniques for real-world time-dependent data.

Psychological Emotion-Based Classification of EEG Variations (Data Analytics) *Oct 2021 – Nov 2022*
Technologies: Neuroscience, EEG Signals, MATLAB, Big Data
Conducted EEG signal analysis to classify emotional states by identifying optimal threshold values. Identical patterns extracted from the EEG bands for emotion recognition.

Product Vending Machine with Machine Learning (IoT Project) *Mar 2022 – May 2022*
Technologies: Arduino, Sensor Technology, IoT, Machine Learning
Built a smart vending machine using Arduino and IoT sensors with RFID authentication. ML was used to analyze seasonal sales data and recommend top-selling products.

Medicare for e-Health Data Management Technologies (Android App) *Feb 2021 – Jun 2021*
Technologies: Android Development, Room DB, Firebase, MVVM
Designed and developed an Android app to manage patient treatment records. Implemented local storage with Room DB and cloud sync with Firebase for real-time access.

Certifications

NPTEL Online Certification	*Big Data Computing*
ICIWRM 2022	*Int. Conf. on Integrated Water Resources Management*
Udemy	*The Data Science Course 2022: Complete Data Science Bootcamp*
PSG NI Virtual Instrumentation Center	*Workshop on Real-Time Applications using LabVIEW with Arduino and Raspberry Pi*
edX	*Building Modern Node.js Applications on AWS*
DevOps Course	*CI/CD (Continuous Integration/Delivery) for Beginners*
Microsoft Certification	*Azure Fundamentals (AZ-900)*
Udemy	*Learn to Build Microservices using Spring Boot 3, Spring Cloud, React, Kafka, RabbitMQ, REST API, Docker & IntelliJ IDEA*